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Enhanced Dehydrogenation Properties of MgH_2 with Delaminated $Ti_3C_2T_x$ MXene Produced by Simple Wet Method

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Abstract

Magnesium hydride (MgH_2) is a promising hydrogen storage material due to its high hydrogen capacity. However, pure MgH_2 demonstrates high dehydrogenation temperature and slow reaction kinetics. To overcome these limitations, in this study, delaminated $Ti_3C_2T_x$ MXene was incorporated into MgH_2 using a simple wet mixing process in order to improve dehydrogenation performance. It was found that increasing the MXene content from 1 to 6 wt.% decreased the onset dehydrogenation temperature of the MgH_2-$Ti_3C_2T_x$ composite from 242 to 218 °C, which is significantly lower than that for pure MgH_2 (325 °C). The activation energy for hydrogen desorption decreased from 145 kJ/mol for materials containing 1 wt.% $Ti_3C_2T_x$ to 67–70 kJ/mol for materials containing 3 or 6 wt.%, which is lower than the values reported for ball-milled MXene-MgH_2 composites. The wet mixing process preserves the layered structure of MXene and offers a potentially scalable approach for industrial hydrogen storage applications.

Keywords: MXene; magnesium hydride; dehydrogenation; composites

1. Introduction

The development of environmentally friendly energy sources plays a crucial role in the face of climate change. Hydrogen is considered one of the most promising energy sources due to its high energy density, its environmental friendliness, and its potential for integration into various energy generation systems [1,2].

However, for the widespread implementation of hydrogen energy, a number of technical and economic challenges remain unresolved, the primary one being the safe, efficient, and reversible storage of hydrogen [3,4]. Traditional storage methods in compressed or liquefied forms have significant drawbacks: high energy consumption, low storage density, and safety risks [5,6]. Therefore, the focus is shifting to solid-state storage materials capable of absorbing and releasing hydrogen under moderate conditions. Among such materials, magnesium hydride (MgH_2) stands out for its high hydrogen storage capacity (~7.6 wt.%), good reversibility, and relatively low cost [7–9]. The main obstacles to its practical application remain the high onset temperature of more than 300 °C that is required for

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hydrogen release and slow reaction kinetics, which necessitates finding ways to improve these characteristics [10,11].

One of the most proposed approaches to modifying the dehydrogenation properties of MgH_2 is the addition of various catalysts [12,13]. Recently, particular interest has been generated by the use of two-dimensional MXene materials, specifically $Ti_3C_2T_x$, which possess unique physicochemical properties, such as high electrical conductivity, high surface area, and the ability for interfacial interaction [14,15]. $Ti_3C_2T_x$ has demonstrated potential as a catalyst for accelerating the hydrogenation and dehydrogenation processes in metal hydrides [16–18]. The addition of $Ti_3C_2T_x$ to MgH_2 can significantly reduce the dehydrogenation temperature and improve hydrogen storage kinetics. $Ti_3C_2T_x$'s surface-active groups (e.g., -OH, -F, -O) provide sorption sites for hydrogen molecules and facilitate their dissociation into hydrogen atoms [19,20]. Simultaneously, its high electrical conductivity accelerates electron transfer within the material [21]. This not only further promotes hydrogen dissociation and diffusion, but also significantly reduces the reaction activation energy for hydrogen sorption and desorption in magnesium hydride [22]. These combined effects enhance the dehydrogenation reaction kinetics, thereby lowering the dehydrogenation temperature to $\leq$250 °C [16,23]. Moreover, the incorporation of MXene also improves the kinetics of the hydrogenation process, leading to an overall enhancement in hydrogen storage performance [24].

Adding $Ti_3C_2T_x$ to MgH_2 can prevent particle agglomeration and surface oxidation, which are the main causes of decreased hydrogen storage efficiency after multiple hydrogenation/dehydrogenation cycles. The two-dimensional structure of $Ti_3C_2T_x$ forms a protective layer on the surface of MgH_2 particles, inhibiting their growth and agglomeration, as well as enhancing the chemical stability of the material. Experimental data show that MgH_2–$Ti_3C_2T_x$ composites retain high hydrogen storage capacity after several cycles, demonstrating minimal hydrogen capacity loss. This indicates the high stability of such composites under cyclic operating conditions [25,26].

Liu et al. [23] first introduced Ti_3C_2 MXene into the MgH_2 system and found that the addition of 5 wt.% Ti_3C_2 reduced the onset dehydrogenation temperature to 185 °C, which is 93 °C lower than that of pure MgH_2 (278 °C), and lowered the activation energy to 98.9 kJ/mol, demonstrating the excellent catalytic effect of Ti_3C_2 on MgH_2. Wu et al. [27] further employed multilayer Ti_3C_2 as a catalyst for MgH_2, achieving a lower onset dehydrogenation temperature of 142 °C with 6 wt.% loading, while the activation energy was 99.0 kJ/mol. Subsequent studies revealed that the catalytic activity of $Ti_3C_2T_x$ can be further enhanced by controlling the exposed active facets. For instance, MgH_2 composites containing 5 wt.% edge-facet-rich $Ti_3C_2T_x$ achieved an onset temperature of 199 °C and an activation energy of 78.2 kJ/mol [17]. Moreover, bimetallic MXenes such as Ti_2VC_2 exhibit superior catalytic performance due to the synergistic effect between different transition metals, with MgH_2 composites containing 10 wt.% Ti_2VC_2 exhibiting an onset temperature as low as 170 °C and an activation energy of 71 kJ/mol [28].

In most studies, MXene was mixed with MgH_2 by ball milling. Although ball milling is effective in refining particles and surface activation, the high-energy impact and shear forces involved can cause fracture or chemical reaction of the two-dimensional MXene sheets, thereby degrading the layered morphology and causing the change in surface functional groups [29]. Such structural degradation reduces the exposed edge sites and surface terminations, thereby weakening the catalytic effect of MXene on MgH_2 dehydrogenation [21]. In addition, ball milling is energy-intensive and sensitive to milling parameters, which poses challenges for reproducibility and scalability [30].

In contrast, the wet mixing process adopted in this work offers a gentler alternative. By avoiding high-energy milling, this liquid-phase route can preserve both the layered

morphology and surface chemistry of $Ti_3C_2T_x$, while also achieving its uniform dispersion on MgH_2 particles. The intimate interfacial contact in solution is expected to facilitate catalytic dehydrogenation, thereby improving dehydrogenation properties.

Moreover, most previous studies used relatively high MXene loadings (5 wt.% or more), and the catalytic behavior of $Ti_3C_2T_x$ at lower loadings remains underexplored. To address this gap, in the present work, MgH_2–delaminated $Ti_3C_2T_x$ composites with different MXene contents (1, 3, and 6 wt.%) were prepared by a wet mixing method. The effect of MXene content on the dehydrogenation temperature and activation energy was investigated, aiming to provide experimental evidence for the development of efficient catalysts at low loadings.

2. Materials and Methods

The pristine MgH_2 powder (average particle size 40–50 μm) was obtained by hydrogenation of magnesium powder (MPF 4, 99.2% purity, produced in Russia).

For the synthesis of the $Ti_3C_2T_x$ material, a powder of the Ti_3AlC_2 MAX-phase with the average particle size of up to 40 μm (Wuhan Golden Kylin Industry & Trade Co., Ltd., Wuhan, China) was used.

First, an etching solution was prepared by mixing 11.65 mL of deionized water, 24 mL of hydrochloric acid (HCl, 37%), and 4.35 mL of hydrofluoric acid (HF, 45%), corresponding to volume percentages of 29% H_2O, 60% HCl, and 11% HF. The Ti_3AlC_2 (2 g) powder was placed in the prepared acid solution and kept under stirring at 200 rpm for 24 h to remove the aluminum atomic layers from the MAX phase. After etching, the suspension was centrifuged (3500 rpm, 5 min) to separate the solid product from the acidic supernatant. The precipitate was then washed with deionized water, and the centrifugation was repeated. This cycle was repeated until the pH of the supernatant reached 6–7. Next, 50 mL of 1M LiCl solution was added to the precipitate and stirred for 24 h for lithium intercalation. Following intercalation, the aqueous $Ti_3C_2T_x$–LiCl dispersion was washed by centrifugation at 3500 rpm to remove excess LiCl. Washing was continued until a dark, non-sedimenting dispersion of delaminated $Ti_3C_2T_x$ was obtained. The resulting supernatant was then subjected to several washing cycles by centrifugation at 10,000 rpm, with water progressively replaced by isopropyl alcohol during each cycle until complete solvent exchange was achieved. Finally, the sediment containing delaminated $Ti_3C_2T_x$ was redispersed in anhydrous isopropyl alcohol. The resulting $Ti_3C_2T_x$ material was added to MgH_2. To improve dispersion and contact between the components, a wet mixing method in an alcoholic medium was used. MgH_2 powder was mixed with the synthesized $Ti_3C_2T_x$ (1, 3, and 6 wt.%) in anhydrous alcohol, then stirred at 200 rpm on a magnetic stirrer. After mixing, the suspension was dried under vacuum (approximately −0.1 MPa) at 120 °C for 6 h to completely remove the solvent. Figure 1 schematically illustrates the preparation process of MgH_2–delaminated $Ti_3C_2T_x$ composites via the wet mixing method.

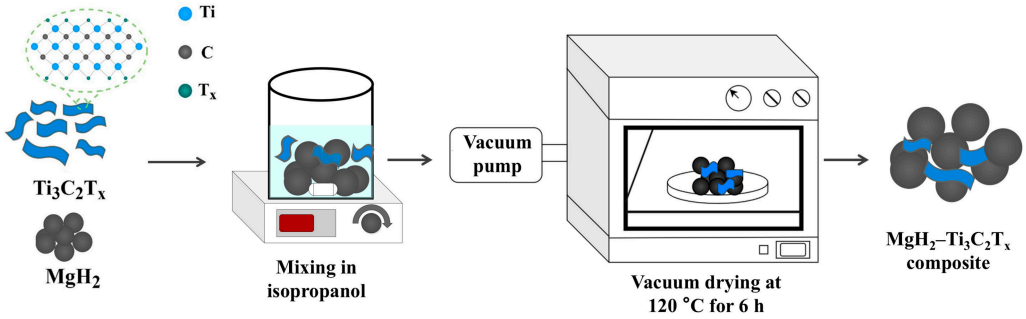

Figure 1. Schematic illustration of the preparation process of MgH_2–delaminated $Ti_3C_2T_x$ composites via the wet mixing method.

The phase composition of the MgH_2–$Ti_3C_2T_x$ composites was studied using an XRD-7000 diffractometer (Shimadzu, Kyoto, Japan). The measurements were performed using Cu Kα radiation (λ = 1.5406 Å) at 40 kV and 30 mA. Sample scanning was conducted in Theta-2Theta mode over an angular range of 2θ from 3.5° to 90°. The scan rate was 10°/min with a step size of 0.0143°. The signal accumulation time at each point was 21.49 s. The morphology of the composite samples was studied by scanning electron microscopy using VEGA 3 SBU (TESCAN, Brno, Czech Republic) equipped with an OXFORD X-Max 50 energy-dispersive spectrometer. Prior to analysis, the samples were mounted on conductive carbon tape. The accelerating voltage was 20 kV. The study of the specific surface area of the samples was performed using the low-temperature nitrogen adsorption method on a Sync 420A analyzer (3P Instruments, Odelzhausen, Germany). TEM and STEM analyses were performed on a JEM-2100F (HR) microscope (JEOL, Tokyo, Japan) operated at an acceleration voltage of 200 kV. The STEM image was acquired in dark-field mode at a magnification of 30,000×. The HRTEM image was acquired at a magnification of 600,000. X-ray photoelectron spectroscopy (XPS) measurements were performed on a spectrometer (SPECS GmbH, Berlin, Germany) equipped with a PHOIBOS-100-MCD-5 hemispherical analyzer using non-monochromatized Al Kα radiation (hν = 1486.6 eV). The binding energy scale was calibrated using the Au $4f_{7/2}$ (84.0 eV) and Cu $2p_{3/2}$ (932.7 eV) lines. Charge correction was performed using the C 1s line of adventitious carbon (284.5 eV) as an internal reference. Survey spectra and high-resolution spectra of Al 2p, C 1s, O 1s, Ti 2p, Mg 1s, F 1s, Cl 2p, and Mg 2p were recorded at a pass energy of 20 eV. The relative surface elemental composition was determined from the integrated intensities of the photoelectron lines, corrected by the corresponding atomic sensitivity factors. Prior to measurements, the samples were degassed at 200 °C for 10 h under vacuum to clean the surface. Measurements were carried out at the boiling point of nitrogen (77 K) in the relative pressure range of P/P0 = 0.005–0.995. Data from the relative pressure region of 0.05–0.30 were used to calculate the specific surface area using the BET method. To investigate the temperature and kinetics of hydrogen desorption from the MgH_2–$Ti_3C_2T_x$ composite, the method of thermal desorption spectroscopy (TDS) was employed using the Gas Reaction Automated Machine (GRAM, Tomsk Polytechnic University, Tomsk, Russia) developed by Tomsk Polytechnic University and equipped with an RGA100 mass spectrometer. The measurements were performed under vacuum (approximately 10^{-2}–10^{-3} Pa) from room temperature to 500 °C at a heating rate of 4, 6, and 8 °C/min. Hydrogen sorption properties were determined by measuring the pressure–composition isotherms (PCIs) using the GRC apparatus. The sample was placed inside the chamber and evacuated to a base pressure lower 10^{-3} Pa. Then, the sample was heated to 450 °C with a heating rate of 10 °C/min. The PCI was performed over a hydrogen pressure range of 0–30 atm, during which the absorption and desorption kinetic curves were recorded. After each desorption, the sample was cooled under vacuum. This procedure was repeated for five cycles. To determine the absolute hydrogen concentration in the composites, the RHEN602 (LECO, St. Joseph, MI, USA) hydrogen analyzer was used.

3. Results and Discussions

Figure 2 shows the morphology and crystalline structure of delaminated $Ti_3C_2T_x$ and pristine MgH_2. The TEM image of delaminated $Ti_3C_2T_x$ (Figure 2a) reveals a sheet-like morphology, and the cross-sectional TEM image (Figure 2b) shows a well-stacked lamellar structure. The XRD pattern of $Ti_3C_2T_x$ (Figure 2c) exhibits a clear (002) peak at 2θ = 6.76°, corresponding to an interlayer spacing of approximately 1.31 nm, while that of pristine MgH_2 (Figure 2e) matches the tetragonal MgH_2 phase.

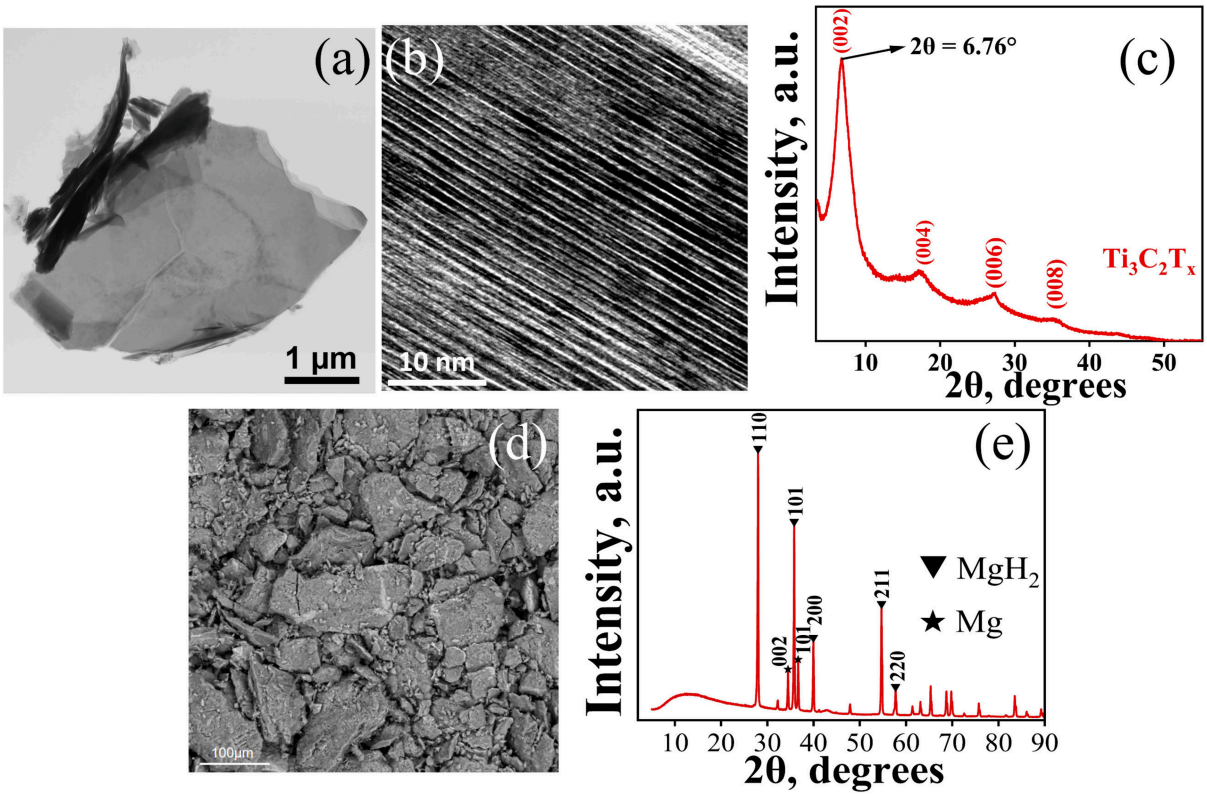

Figure 2. (**a**) TEM image and (**b**) cross-sectional TEM image of delaminated $Ti_3C_2T_x$; (**c**) XRD pattern of delaminated $Ti_3C_2T_x$; (**d**) SEM image and (**e**) XRD pattern of the pristine MgH_2 powder.

Figure 3 shows SEM images and corresponding EDX elemental maps for the MgH_2–$Ti_3C_2T_x$ composites with varying $Ti_3C_2T_x$ content. All composites exhibit similar morphology. The average particle sizes are 25, 36, and 16 μm for the 1, 3, and 6 wt.% composites, respectively. The absence of a clear trend in average particle size may be attributed to the low MXene content (1–6 wt.%), the gentle nature of the wet mixing process, and the presence of occasional large MgH_2 particles in some samples. For example, the 3 wt.% composite contains a notably large particle (Figure 3b), which increases the average value. Since MgH_2 is the main component and the MXene content is low, the particle size is determined primarily by the starting MgH_2 powder rather than the MXene content.

Further analysis of the EDX elemental maps reveals that at a $Ti_3C_2T_x$ content of 1 wt.%, the Ti signal is weak and dispersed. When the content increases to 3 wt.%, the Ti distribution clearly overlaps with the MgH_2 regions, indicating that most MgH_2 particles are covered by $Ti_3C_2T_x$. At 6 wt.%, the Ti signal is significantly enhanced in the regions of fine MgH_2 particles, while the coverage on larger particles becomes uneven, and flake- or block-like agglomerates appear in some areas (Figure 3c,i), suggesting stacking and aggregation of MXene layers at higher concentrations. These results indicate that the structure of $Ti_3C_2T_x$ and its functional groups remain stable after composite synthesis, but agglomeration should be suppressed at higher MXene loadings.

In general, as the $Ti_3C_2T_x$ content increases, the uniformity of the dispersion deteriorates: from a nearly uniform distribution to the formation of localized aggregation zones. In this study, local agglomeration occurs at 6 wt.%.

In contrast, most studies on MgH_2–MXene composites focus on catalytic properties rather than dispersion uniformity [26,27], and the effect of MXene content on dispersion homogeneity is rarely discussed. The wet mixing method used in this work reveals a dispersion limit of around 6 wt.% for this method. Notably, the self-assembly approach

enables uniform dispersion at much higher MXene loadings (up to 40 wt.%) via F–Mg chemical bonding, but requires complex processing conditions [31].

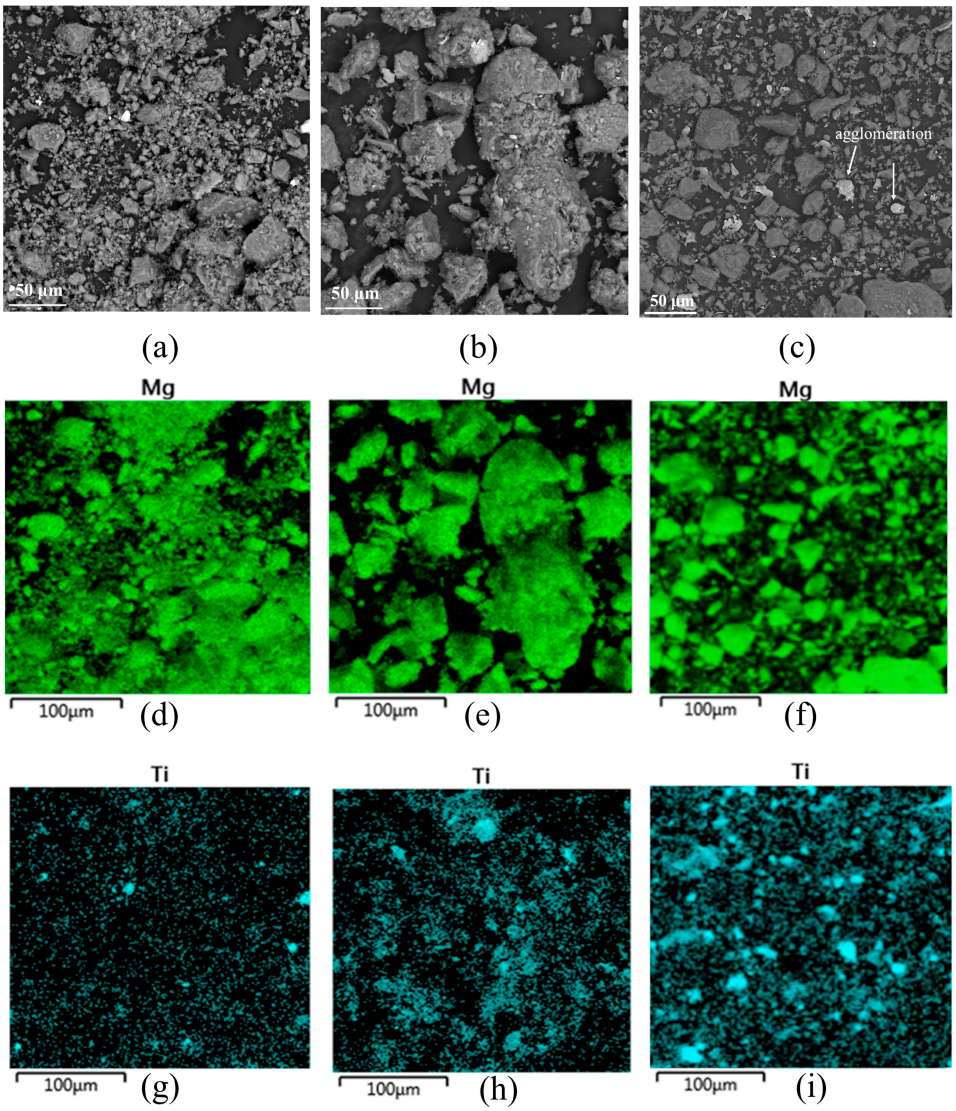

Figure 3. SEM images with the corresponding EDX maps of Mg and Ti for MgH_2–$Ti_3C_2T_x$ composites: 1 wt.% $Ti_3C_2T_x$ (**a**,**d**,**g**); 3 wt.% $Ti_3C_2T_x$ (**b**,**e**,**h**); 6 wt.% $Ti_3C_2T_x$ (**c**,**f**,**i**).

Figure 4 shows XRD patterns of the MgH_2-based composites with different contents of $Ti_3C_2T_x$. The introduction of MXene into the composite does not change the structure of the magnesium hydride, which exhibits characteristic peaks corresponding to the tetragonal structure of MgH_2. A characteristic diffraction peak of the (002) basal plane of $Ti_3C_2T_x$ MXene can be seen in Figure 4b. The intensity of this peak increases with increasing MXene content from 1 to 3 wt.%, but shows a slight decrease at 6 wt.%. No peak was detected for the sample containing 1 wt.% $Ti_3C_2T_x$.

The lower intensity of the (002) diffraction peak at 6 wt.% could be related to the local agglomeration observed in the SEM images (Figure 3c,i), which might lead to partial stacking or disordering of the MXene layers at higher concentrations. However, further characterization would be needed to confirm this interpretation.

To further investigate the surface chemical state and interfacial interactions between $Ti_3C_2T_x$ and MgH_2, XPS analysis was performed on both pristine $Ti_3C_2T_x$ and the MgH_2–6 wt.% $Ti_3C_2T_x$ composite (Figure 5). The C 1s spectrum of $Ti_3C_2T_x$ shows four components at 282.0, 284.5, 288.2, and 289.8 eV. The components at 282.0 and 284.5 eV, present

in both samples, are assigned to C–Ti (MXene framework) and C–C (adventitious carbon), respectively [32]. The component at 288.2 eV, observed only in $Ti_3C_2T_x$, is attributed to oxygen-containing carbon species (C=O/O–C=O). The component at 289.8 eV corresponds to the Mg KLL Auger line rather than a carbon species, which overlaps the C 1s binding energy region.

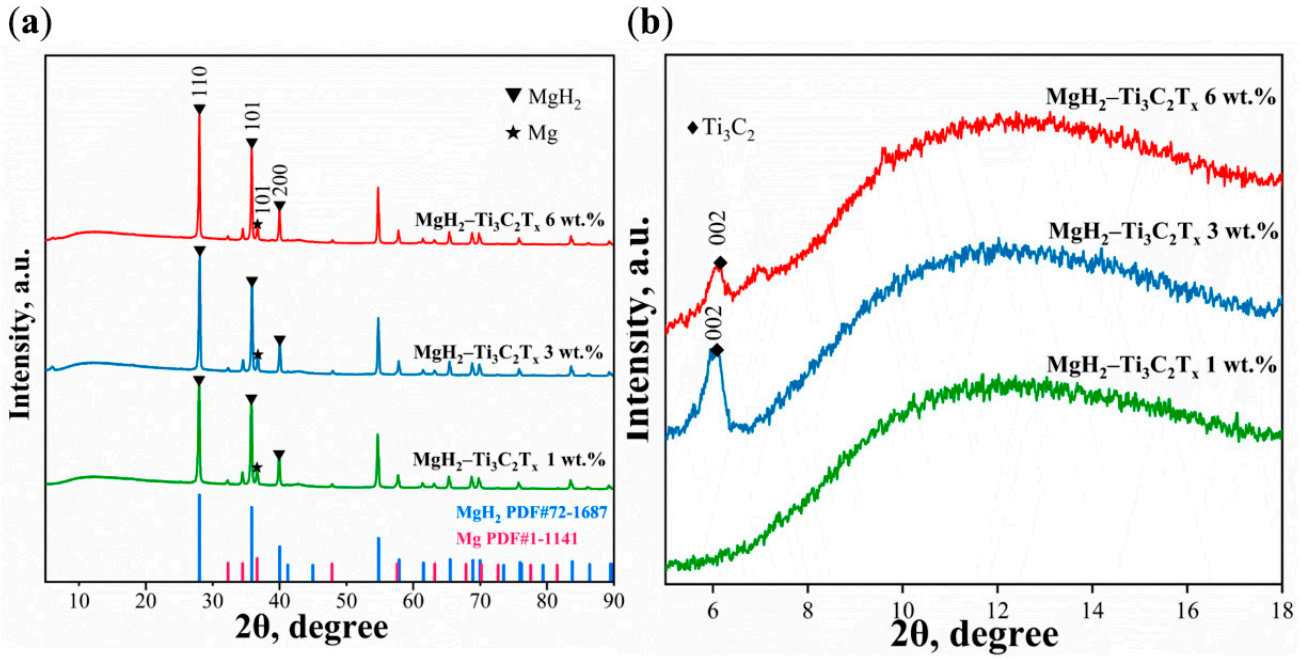

Figure 4. X-ray diffraction patterns of MgH_2–$Ti_3C_2T_x$ composites with different $Ti_3C_2T_x$ contents (**a**); enlarged XRD pattern at 2θ range of 3.5–19° (**b**).

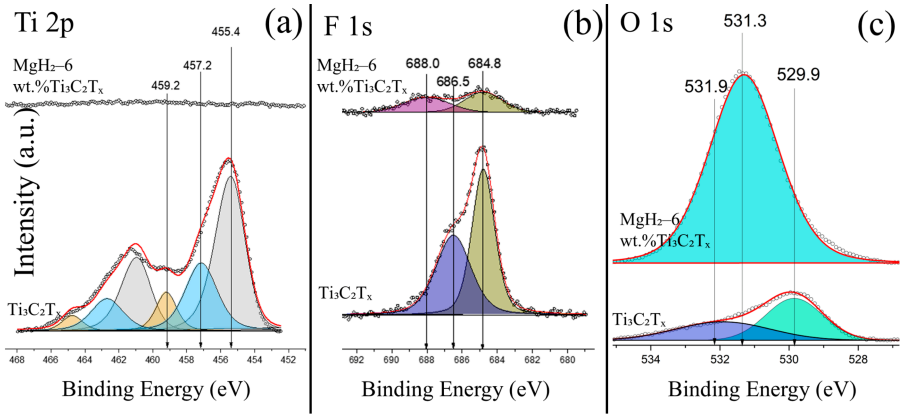

Figure 5. XPS spectra of $Ti_3C_2T_x$ and MgH_2–6 wt.% $Ti_3C_2T_x$. (**a**) Ti 2p, (**b**) F 1s, and (**c**) O 1s regions.

The Ti 2p spectrum of $Ti_3C_2T_x$ shows three main components at 455.4, 457.2, and 459.2 eV (Figure 5a). The component at 455.4 eV corresponds to Ti–C with possible contributions from Ti^{2+} and Ti–O/Ti–OH terminations [32]. The component at 457.2 eV is assigned to low-valent Ti^{3+}, consistent with previous reports [32]. The component at 459.2 eV is attributed to titanium oxyfluoride ($TiO_{2-x}F_x$), indicating the coexistence of –O and –F terminations [27,32].

The F 1s spectrum of $Ti_3C_2T_x$ (Figure 5b) shows two components at 684.8 and 686.5 eV, assigned to C–Ti–F_x (–F terminations) and $TiO_{2-x}F_x$, respectively [32]. After composite formation, the relative intensity of the C–Ti–F_x component decreases, while a new compo-

nent at 688.0 eV appears. This new component is attributed to interfacial fluorine species, possibly involving F in a mixed Ti/Mg environment, or surface organic fluorine.

The Cl 2p spectrum of $Ti_3C_2T_x$ exhibits a doublet at 199.1 eV, assigned to Ti–Cl terminations [33]. This signal disappears in the composite, suggesting that –Cl terminations are removed or replaced during composite formation or their content is under the detection limit of XPS.

The Mg 2p spectrum of the composite shows a single peak at 49.4 eV, consistent with Mg^{2+} [34]. The binding energy is close to both MgH_2 (~50.1 eV) and MgO (~50.5 eV), so a definitive assignment between these species cannot be made from Mg 2p alone.

The O 1s spectrum of $Ti_3C_2T_x$ exhibits two components at 529.9 and 531.9 eV (Figure 5c). The component at 529.9 eV is assigned to Ti–O species, which may originate from –O terminations or from surface TiO_2. Although no TiO_2 crystalline phase was detected by XRD, its presence cannot be completely excluded. The component at 531.9 eV is assigned to –OH terminations, with a possible contribution from adsorbed water [32].

In the composite, the O 1s spectrum is dominated by a component at 531.3 eV, assigned to Mg–O [34]. This signal may originate from partial oxidation of MgH_2 to $MgO/Mg(OH)_2$ during sample handling, or from interfacial interaction between Mg and the –O terminations of MXene. These two contributions cannot be distinguished from O 1s alone.

Taken together, the disappearance of Ti signals in the composite (observed in both Ti 2p and O 1s spectra), together with the changes in terminal-group signals observed in F 1s and Cl 2p, indicate that the MXene surface is most likely covered by MgH_2. Since $Ti_3C_2T_x$ is confirmed in the bulk by XRD and SEM-EDS, the absence of Ti signals in XPS is attributed to surface coverage rather than insufficient MXene content. The chemical environment of the terminal groups (–O, –F, –Cl) is modified, suggesting their possible participation in interfacial interactions.

To evaluate the effect of the $Ti_3C_2T_x$ content on the specific surface area of the composites, BET analysis was performed. Table 1 shows the specific surface area, pore volume, and average pore diameter for pure MgH_2 and composites with different MXene contents.

Table 1. XPS peak positions and assignments for $Ti_3C_2T_x$ and MgH_2–6 wt.% $Ti_3C_2T_x$.

Region	BE (eV)	Assignment	Sample	Ref.
Ti 2p	455.4	Ti–C (Ti^{2+}, Ti–O/Ti–OH)	$Ti_3C_2T_x$	[32]
	457.2	Ti^{3+}	$Ti_3C_2T_x$	[32]
	459.2	$TiO_{2-x}F_x$	$Ti_3C_2T_x$	[32]
C 1s	282.0	C–Ti	Both	[32]
	284.5	C–C	Both	[32]
	288.2	C=O/O–C=O	$Ti_3C_2T_x$	[27]
	289.8	Mg KLL Auger	Composite	-
O 1s	529.9	C–Ti–O (–O)	$Ti_3C_2T_x$	[32]
	531.9	C–Ti–OH (–OH)	$Ti_3C_2T_x$	[32]
	531.3	Mg–O	Composite	[34]
F 1s	684.8	C–Ti–F_x	Both	[32]
	686.5	$TiO_{2-x}F_x$	Both	[32]
	688.0	Interfacial F	Composite	-
Cl 2p	199.1	Ti–Cl	$Ti_3C_2T_x$	[32,34]
Mg 2p	49.4	Mg^{2+}	Composite	[34]

As shown in Table 2, the addition of $Ti_3C_2T_x$ leads to a decrease in both pore volume and specific surface area. This may be due to the microporous structure of MgH_2 being covered or filled by catalyst particles. A similar decrease in BET surface area and pore volume upon catalyst addition has been reported in MgH_2-based systems, e.g., with mischmetal as a catalyst [35]. Nevertheless, the dehydrogenation properties of the MgH_2–$Ti_3C_2T_x$ composites are improved with increasing MXene content, indicating that the increase in specific surface area is not the primary factor responsible for the reduction in dehydrogenation temperature. Figure 6 presents the temperature programmed desorption curves for pure MgH_2 and MgH_2 samples with 1, 3, and 6 wt.% $Ti_3C_2T_x$. As a representative example, the onset temperatures are considered for a heating rate of 6 °C/min. The onset dehydrogenation temperature was determined as the temperature at which the hydrogen signal first deviated from the baseline in the TDS curves. The onset dehydrogenation temperature of pure MgH_2 is 325 °C. With increasing MXene content, the onset temperature decreases to 242 °C, 231 °C, and 218 °C for the 1, 3, and 6 wt.% samples, respectively, corresponding to reductions of 83 °C, 94 °C, and 107 °C.

Table 2. BET analysis data.

Parameter	BET Specific Surface Area, m^2/g	Total Pore Volume, cm^3/g	Average Pore Size, nm
MgH_2	9.4	0.075	8.2
$Ti_3C_2T_x$	6.6	0.047	28.2
MgH_2–$Ti_3C_2T_x$ (1 wt.%)	7.8	0.017	7.7
MgH_2–$Ti_3C_2T_x$ (3 wt.%)	8.1	0.017	8.5
MgH_2–$Ti_3C_2T_x$ (6 wt.%)	8.8	0.012	8.7

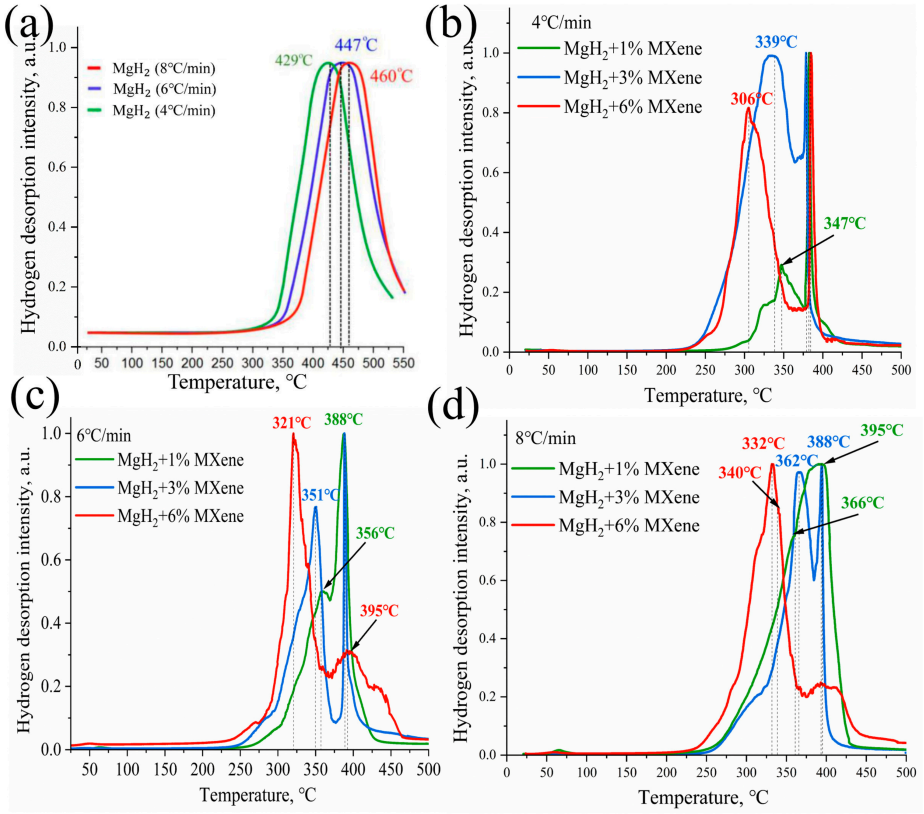

Figure 6. TPD curves of MgH_2 (**a**) and MgH_2 with 1, 3, and 6 wt.% $Ti_3C_2T_x$ at heating rates of 4 °C/min (**b**), 6 °C/min (**c**) and 8 °C/min (**d**).

Compared with previously reported MgH_2–$Ti_3C_2T_x$ composites prepared by ball milling, Wu et al. [27] achieved an onset temperature of 142 °C (a reduction of 125 °C) with 6 wt.% ML-Ti_3C_2, while Gao et al. [17] reported an onset temperature of 199 °C (a reduction of ~101 °C) with 5 wt.% F-$Ti_3C_2T_x$. In the present study, it was possible to reduce the temperature by 107 °C for the materials containing 6 wt.% $Ti_3C_2T_x$, by using a relatively simple wet mixing method at ambient temperature, without the need for high-energy ball milling. Although the absolute onset temperature in this work (218 °C) is higher than in those studies, the reduction is comparable to that of Gao et al. and only slightly lower than that of Wu et al., while offering a simpler and more scalable preparation route. Among the investigated ratios, the lowest onset temperature is observed at 6 wt.% $Ti_3C_2T_x$, confirming the catalytic effect of $Ti_3C_2T_x$ on the dehydrogenation kinetics of MgH_2.

As the MXene content increases, the main desorption peak shifts toward lower temperatures, indicating that MXene effectively catalyzes hydrogen desorption processes and improves the kinetics. In all three composites, two hydrogen desorption peaks are observed on the thermal desorption curves.

The two-peak behavior may have several possible explanations. First, the intrinsic multi-step decomposition of MgH_2 could produce multiple peaks; however, pure MgH_2 exhibits only a single desorption peak under identical conditions (Figure 5a), which rules out this possibility. Second, the γ/β phase transition of MgH_2 has been reported to cause double peaks in ball-milled samples [36]. In the present work, however, no γ-MgH_2 phase was detected by XRD; its characteristic peaks at $2\theta \approx 28.5°$ and 32.5° are absent from the XRD patterns, and the wet mixing process does not involve the high-energy milling required to induce such a transition. Therefore, the two-peak behavior is most likely related to the non-uniform coverage of individual MgH_2 particles by MXene nanosheets. The EDX maps show no large-scale local enrichment or depletion of MXene, indicating that this heterogeneity is not a macroscopic compositional fluctuation but rather a local difference in MXene coverage at the particle scale. The first desorption peak is attributed to the catalytic effect of $Ti_3C_2T_x$, whereas the second is likely due to the fact that the MgH_2 surface is not fully encapsulated by the additive. Notably, even the temperature of the second peak (378–395 °C) is lower than that of pure MgH_2 (429–460 °C), suggesting that even regions not directly covered by MXene may still benefit from partial catalytic effects.

The terminal groups (–F and –O) on the $Ti_3C_2T_x$ MXene surface and the formed $TiO_{2-x}F_x$ phase act synergistically to enhance the dehydrogenation performance of MgH_2. It was found that the introduction of 5 wt.% $TiO_{2-x}F_x$ can lower the onset desorption temperature from 416 °C (pristine MgH_2) to 189 °C and reduce the apparent activation energy from 268 to 120 kJ mol^{-1} [37]. In that case, multivalent Ti species ($Ti^{2+/3+}$) and oxygen vacancies generated in $TiO_{2-x}F_x$ can accelerate electron transfer and facilitate hydrogen diffusion. The –F and –O terminal groups on $Ti_3C_2T_x$ can weaken the Mg–H bond, thereby lowering the barrier and promoting hydrogen recombination. Therefore, it is suggested that the role of the formed $TiO_{2-x}F_x$ and terminal groups collectively contributes to the improved desorption properties of the obtained MXene/MgH_2 composite.

In the present work, the low-temperature peak is attributed to MgH_2 regions in direct contact with $Ti_3C_2T_x$, where the catalyst facilitates dehydrogenation, and the high-temperature peak corresponds to regions with insufficient catalytic coverage. This interpretation is supported by the increase in the low-temperature peak area with increasing MXene content (from 1 to 6 wt.%), indicating more complete coverage of the MgH_2 surface by $Ti_3C_2T_x$. Furthermore, at the same MXene content, the increase in heating rate leads to a shift of the desorption peak towards higher temperatures. This is related to limitations in heat transfer and the specific kinetics of thermal processes [31]. Nevertheless, regardless

of the heating rate, increasing the $Ti_3C_2T_x$ content consistently promotes a decrease in desorption temperature and enhances the catalytic effect.

To evaluate the efficiency of hydrogen desorption from the MgH_2–$Ti_3C_2T_x$ composite, an analysis of thermally stimulated hydrogen desorption was conducted using the Kissinger method [38]. Figure 7 shows the plots of $\ln(\beta/T_p^2)$ versus $1000/T_p$ for pure MgH_2 and MgH_2 composites with different $Ti_3C_2T_x$ content. The calculated dehydrogenation activation energy values for pristine MgH_2 are 189 kJ/mol that is consistent with our previous studies [39] and other works [40].

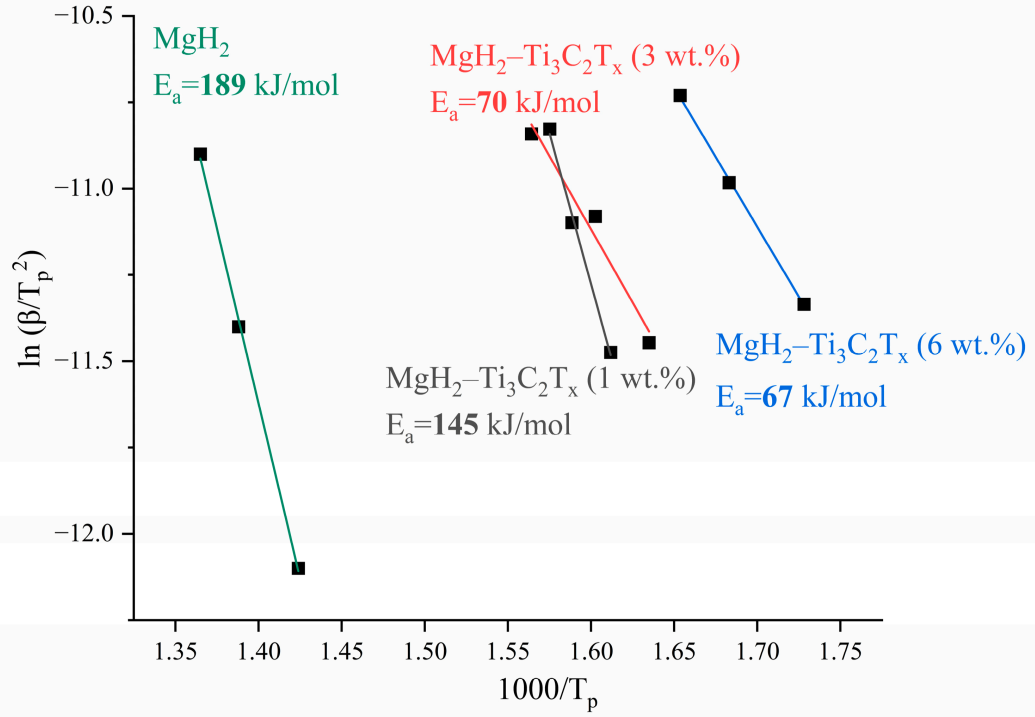

Figure 7. Plots of $\ln(\beta/T_p^2)$ versus $1000/T_p$ for pure MgH_2 and MgH_2 composites with 1, 3, 6 wt.% $Ti_3C_2T_x$.

The activation energy of the first hydrogen desorption maximum was 145, 70, and 67 kJ/mol for 1, 3 and 6 wt.% $Ti_3C_2T_x$, respectively. For comparison, Wu et al. [27] reported an activation energy of 99 kJ/mol for a ball-milled composite containing 6 wt.% ML-Ti_3C_2, while Gao et al. [17] reported a value of 78.2 kJ/mol for a composite with 5 wt.% F-$Ti_3C_2T_x$. In the present work, the activation energies of the 3 and 6 wt.% composites are 70 and 67 kJ/mol, respectively, both lower than the literature values, even lower than the 71 kJ/mol reported by Liu et al. in the bimetallic MXene catalyst system with 10 wt.% Ti_2VC_2 added [28]. This indicates that the wet-mixed $Ti_3C_2T_x$ is highly effective in reducing the kinetic barrier, achieving superior catalytic activity even at lower MXene content and without high-energy ball milling. The calculated activation energies for the second maximum were 192, 152, and 155 kJ/mol, which are close to the dehydrogenation activation energy values for pristine MgH_2. This also confirms the insufficient coverage of magnesium hydride particles at low MXene concentration in the composite.

The activation energy reflects the intrinsic energy barrier of the dehydrogenation reaction, which is effectively reduced by the catalytic $Ti_3C_2T_x$. In addition, the onset temperature is influenced by physical factors such as particle size and defect density. Varin et al. reported that when the MgH_2 particle size was reduced to approximately 500–600 nm by ball milling, the onset dehydrogenation temperature decreased by 40–60 °C, which was attributed to simultaneous particle refinement, defect introduction, and the formation of

the metastable γ-MgH_2 phase [41]. In the present work, the MgH_2 matrix was not subjected to ball milling; therefore, the onset temperature primarily reflects the reduction in the energy barrier due to chemical catalysis. The experimental results show that the addition of $Ti_3C_2T_x$ significantly reduces the dehydrogenation energy barrier of MgH_2, especially at a content of 3 wt.%, where the most pronounced reduction is observed. A further increase in MXene content does not lead to a notable further decrease in the activation energy.

Figure 8 summarizes the dehydrogenation activation energy as a function of the onset desorption temperature for the composites in this work and other MXene-based MgH_2 composites from the literature [16,17,23,27,28,42,43]. The activation energies of the 3 and 6 wt.% composites (70 and 67 kJ/mol) are lower than the literature values, indicating that the wet-mixed $Ti_3C_2T_x$ exhibits excellent catalytic activity even at relatively low MXene loadings. It should be noted that the literature data shown in Figure 8 were obtained under different experimental conditions, including various preparation methods, catalyst loadings and heating rates. Therefore, the comparison provides a general overview of the catalytic effect of MXene rather than a strict quantitative benchmark.

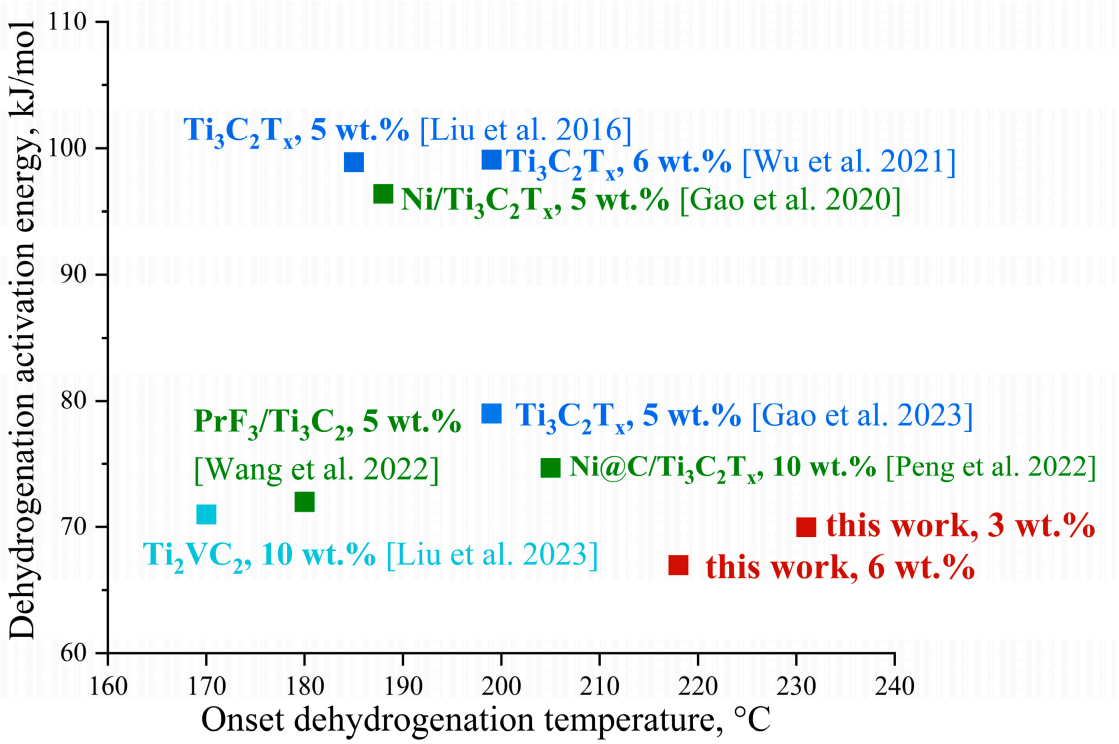

Figure 8. Dehydrogenation activation energy as a function of onset desorption temperature for $Ti_3C_2T_x$ composites (this work) and other MXene-based catalysts from the literature [16,17,23,27,28,42,43].

Figure 9 shows the PCI curves for five cycles. The sorption and desorption curves of all five cycles exhibit similar shapes, and the slopes of the sorption curves are nearly identical across cycles. This indicates that the residual hydrogen content at the start of each cycle was similar and that the sorption and desorption kinetics remained stable during cycling. The measured absolute hydrogen concentration in the composite after five cycles was ~0.5 wt.%. Thus, the effective (reversible) hydrogen capacity of the obtained composites was approx. 6.4 wt.%. It should be noted that only five cycles were performed in this work, and therefore longer cycling would be needed to fully assess the long-term degradation of the composite.

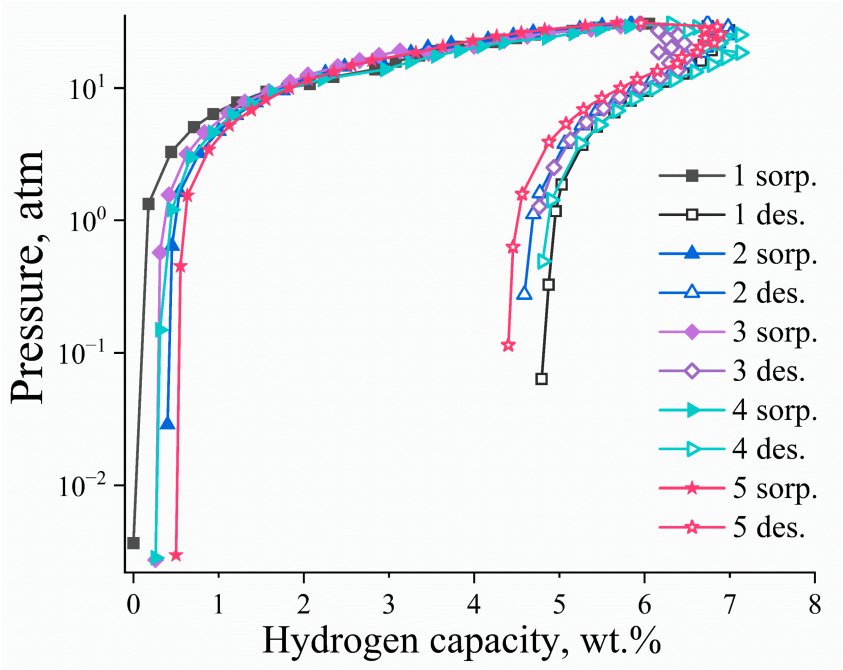

Figure 9. Pressure–composition isotherms (PCI) of the MgH_2–$Ti_3C_2T_x$ composite over five absorption–desorption cycles.

4. Conclusions

In the present work, MgH_2–$Ti_3C_2T_x$ composites containing 1, 3, and 6 wt% MXene were obtained using the wet mixing method. The wet mixing method offers a simple, scalable route for preparing MXene-catalyzed MgH_2 composites while preserving the layered structure of MXene. MXene nanosheets were uniformly distributed in the MgH_2 matrix at concentrations of 1 and 3 wt.%, whereas at 6 wt.% local agglomeration was observed, indicating a dispersion limit of approximately 6 wt.% in this case.

The addition of $Ti_3C_2T_x$ significantly improved the dehydrogenation kinetics. The onset dehydrogenation temperature decreased from 325 °C for pure MgH_2 to 242 °C, 231 °C, and 218 °C for the 1, 3, and 6 wt.% composites, respectively. Pure MgH_2 exhibits a single desorption peak (429–460 °C, depending on the heating rate) with an activation energy of 189 kJ/mol. In contrast, all composites show two distinct desorption peaks: a low-temperature peak (306–366 °C) attributed to MXene-catalyzed MgH_2, and a high-temperature peak (378–395 °C) corresponding to uncatalyzed regions. The activation energy of the low-temperature peak decreases significantly with increasing MXene content, from 145 kJ/mol at 1 wt.% to 70 and 67 kJ/mol at 3 and 6 wt.%, confirming the catalytic effect of $Ti_3C_2T_x$.

Based on XPS analysis it is suggested that the synergetic effect of MXene terminal groups (mostly -O, -F) and the formed $TiO_{2-x}F_x$ is responsible for enhanced dehydrogenation properties of the MXene/MgH_2 composites.

Cycling tests over five sorption/desorption cycles showed stable kinetics, with approximately 0.5 wt.% of irreversibly retained hydrogen and a reversible hydrogen capacity of approximately 6.4 wt.%.

In future work, longer cycling tests will be performed to assess the long-term stability and capacity retention of the composite, and more in-depth investigations will be made to reveal the catalytic mechanism.

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E.K.; funding acquisition, E.K. All authors have read and agreed to the published version of the manuscript.

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